STUDENT ANALYSIS

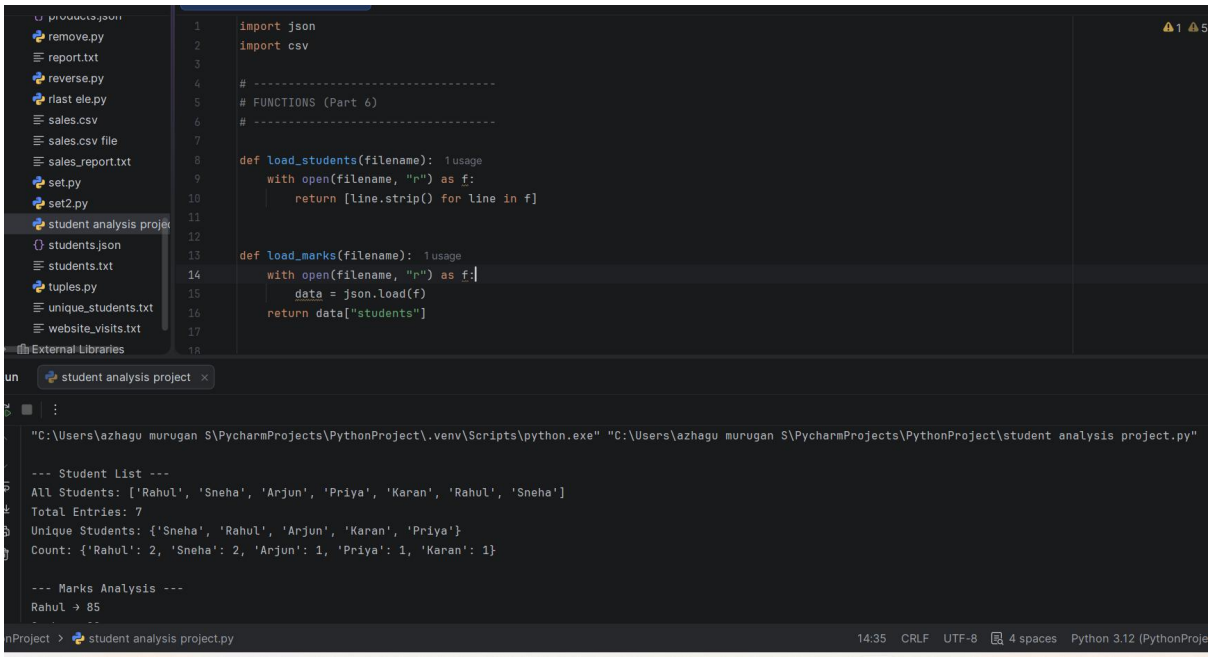
```python
import json
import csv

# ------------------------------------
# FUNCTIONS
# ------------------------------------

def load_students(filename):
    with open(filename, "r") as f:
        return [line.strip() for line in f]


def load_marks(filename):
    with open(filename, "r") as f:
        data = json.load(f)
    return data["students"]


def load_attendance(filename):
    attendance = {}
    with open(filename, "r") as f:
        reader = csv.DictReader(f)
        for row in reader:
            attendance[row["name"]] = {
                "days_present": int(row["days_present"]),
                "total_days": int(row["total_days"])
            }
    return attendance


def calculate_average(marks_list):
    return sum(marks_list) / len(marks_list)


def calculate_percentage(present, total):
    return (present / total) * 100


def get_topper(students):
    return max(students, key=lambda x: x["marks"])


def get_grade(mark):
    if mark >= 90:
        return "A"
    elif mark >= 75:
        return "B"
    elif mark >= 50:
        return "C"
    else:
        return "Fail"


# ------------------------------------
# LOAD DATA
# ------------------------------------
```

```python
students_list = load_students("students.txt")
marks_data = load_marks("marks.json")
attendance_data = load_attendance("attendance.csv")

# -----------------------------------
# PART 1 - TXT ANALYSIS
# -----------------------------------

print("\n--- Student List ---")

print("All Students:", students_list)
print("Total Entries:", len(students_list))

unique_students = set(students_list)
print("Unique Students:", unique_students)

count_dict = {}
for s in students_list:
    count_dict[s] = count_dict.get(s, 0) + 1

print("Count:", count_dict)

with open("unique_students.txt", "w") as f:
    for s in unique_students:
        f.write(s + "\n")


# -----------------------------------
# PART 2 - JSON ANALYSIS
# -----------------------------------

print("\n--- Marks Analysis ---")

marks_list = []
course_count = {}

for s in marks_data:
    print(s["name"], "→", s["marks"])
    marks_list.append(s["marks"])

    course = s["course"]
    course_count[course] = course_count.get(course, 0) + 1

topper = get_topper(marks_data)
lowest = min(marks_data, key=lambda x: x["marks"])

print("Topper:", topper)
print("Lowest:", lowest)

avg_marks = calculate_average(marks_list)
print("Average Marks:", avg_marks)

print("Python Students:")
for s in marks_data:
    if s["course"] == "Python":
        print(s["name"])

print("Course Count:", course_count)


# -----------------------------------
```

```python
# PART 3 - CSV ANALYSIS
# ------------------------------------

print("\n--- Attendance Analysis ---")

attendance_percent = {}

for name, data in attendance_data.items():
    percent = calculate_percentage(data["days_present"], data["total_days"])
    attendance_percent[name] = percent

    print(name, "→", percent)

print("Below 80%:")
for name in attendance_percent:
    if attendance_percent[name] < 80:
        print(name)

best_attendance = max(attendance_percent, key=attendance_percent.get)
print("Best Attendance:", best_attendance)


# ------------------------------------
# PART 4 - DATA STRUCTURES
# ------------------------------------

print("\n--- Data Structures ---")

print("Max Marks:", max(marks_list))
print("Min Marks:", min(marks_list))
print("Sum Marks:", sum(marks_list))

courses_tuple = tuple([s["course"] for s in marks_data])
print("Courses Tuple:", courses_tuple)

courses_set = set(courses_tuple)
print("Unique Courses:", courses_set)

marks_dict = {s["name"]: s["marks"] for s in marks_data}
print("Marks Dict:", marks_dict)

attendance_dict = attendance_percent
print("Attendance Dict:", attendance_dict)


# ------------------------------------
# PART 5 - CONDITIONS
# ------------------------------------

print("\n--- Pass/Fail ---")

for name in marks_dict:
    if marks_dict[name] >= 50:
        print(name, "Pass")
    else:
        print(name, "Fail")

print("\n--- Grades ---")

for name in marks_dict:
    print(name, "→", get_grade(marks_dict[name]))
```

```python
print("\n--- High Performers ---")

for name in marks_dict:
    if marks_dict[name] > 80 and attendance_percent[name] > 85:
        print(name)


# -----------------------------------
# PART 7 - COMBINED ANALYSIS
# -----------------------------------

print("\n--- Combined Data ---")

final_data = {}

for s in marks_data:
    name = s["name"]

    final_data[name] = {
        "marks": s["marks"],
        "attendance": attendance_percent[name],
        "course": s["course"],
        "grade": get_grade(s["marks"])
    }

for name, data in final_data.items():
    print(name, data)


# Eligible students
eligible = []
improvement = []

for name, data in final_data.items():
    if data["marks"] >= 75 and data["attendance"] >= 80:
        eligible.append(name)
    else:
        improvement.append(name)


# -----------------------------------
# PART 8 - FILE OUTPUT
# -----------------------------------

with open("report.txt", "w", encoding="utf-8") as f:
    f.write("Student Report\n")
    for name, data in final_data.items():
        f.write(f"{name} - Marks: {data['marks']} - Attendance:
{round(data['attendance'],1)}% - Grade: {data['grade']}\n")


with open("eligible_students.xlsx", "w") as f:
    for name in eligible:
        f.write(name + "\n")




# -----------------------------------
```

```python
# FINAL OUTPUT
# ------------------------------------

print("\n--- Final Result ---")

print("Topper:", topper["name"])
print("Best Attendance:", best_attendance)
print("Average Marks:", round(avg_marks, 1))
print("Eligible Students:", ", ".join(eligible))
print("Students Needing Improvement:", ", ".join(improvement))
```



```python
# ----------------------------------
# PART 1 - TXT ANALYSIS
# ----------------------------------

print("\n--- Student List ---")

print("All Students:", students_list)
print("Total Entries:", len(students_list))

unique_students = set(students_list)
print("Unique Students:", unique_students)

count_dict = {}
for s in students_list:
    count_dict[s] = count_dict.get(s, 0) + 1

print("Count:", count_dict)
```

```
"C:\Users\azhagu murugan S\PycharmProjects\PythonProject\.venv\Scripts\python.exe" "C:\Users\azhagu murugan S\PycharmProjects\PythonProject\student analysis project.py"

--- Student List ---
All Students: ['Rahul', 'Sneha', 'Arjun', 'Priya', 'Karan', 'Rahul', 'Sneha']
Total Entries: 7
Unique Students: {'Sneha', 'Rahul', 'Arjun', 'Karan', 'Priya'}
Count: {'Rahul': 2, 'Sneha': 2, 'Arjun': 1, 'Priya': 1, 'Karan': 1}

--- Marks Analysis ---
Rahul → 85
```

```python
# ----------------------------------
# PART 2 - JSON ANALYSIS
# ----------------------------------

print("\n--- Marks Analysis ---")

marks_list = []
course_count = {}

for s in marks_data:
    print(s["name"], "→", s["marks"])
    marks_list.append(s["marks"])

    course = s["course"]
    course_count[course] = course_count.get(course, 0) + 1

topper = get_topper(marks_data)
lowest = min(marks_data, key=lambda x: x["marks"])
```

```
Count: {'Rahul': 2, 'Sneha': 2, 'Arjun': 1, 'Priya': 1, 'Karan': 1}

--- Marks Analysis ---
Rahul → 85
Sneha → 92
Arjun → 78
Priya → 88
Karan → 70
Topper: {'name': 'Sneha', 'marks': 92, 'course': 'Data Engineering'}
Lowest: {'name': 'Karan', 'marks': 70, 'course': 'Python'}
```

```python

print("\n--- Attendance Analysis ---")

attendance_percent = {}

for name, data in attendance_data.items():
    percent = calculate_percentage(data["days_present"], data["total_days"])
    attendance_percent[name] = percent

    print(name, "->", percent)

print("Below 80%:")
for name in attendance_percent:
    if attendance_percent[name] < 80:
        print(name)

best_attendance = max(attendance_percent, key=attendance_percent.get)
print("Best Attendance:", best_attendance)
```

```
Course Count: {'Python': 3, 'Data Engineering': 1, 'AI': 1}

--- Attendance Analysis ---
Rahul -> 88.0
Sneha -> 96.0
Arjun -> 80.0
Priya -> 92.0
Karan -> 72.0
Below 80%:
Karan
```

```python
# PART 4 - DATA STRUCTURES
# ---------------------------------

print("\n--- Data Structures ---")

print("Max Marks:", max(marks_list))
print("Min Marks:", min(marks_list))
print("Sum Marks:", sum(marks_list))

courses_tuple = tuple([s["course"] for s in marks_data])
print("Courses Tuple:", courses_tuple)

courses_set = set(courses_tuple)
print("Unique Courses:", courses_set)

marks_dict = {s["name"]: s["marks"] for s in marks_data}
print("Marks Dict:", marks_dict)
```

```
--- Data Structures ---
Max Marks: 92
Min Marks: 70
Sum Marks: 413
Courses Tuple: ('Python', 'Data Engineering', 'Python', 'AI', 'Python')
Unique Courses: {'Data Engineering', 'Python', 'AI'}
Marks Dict: {'Rahul': 85, 'Sneha': 92, 'Arjun': 78, 'Priya': 88, 'Karan': 70}
Attendance Dict: {'Rahul': 88.0, 'Sneha': 96.0, 'Arjun': 80.0, 'Priya': 92.0, 'Karan': 72.0}

--- Pass/Fail ---
```

```python
# ----------------------------------

print("\n--- Pass/Fail ---")

for name in marks_dict:
    if marks_dict[name] >= 50:
        print(name, "Pass")
    else:
        print(name, "Fail")

print("\n--- Grades ---")

for name in marks_dict:
    print(name, "→", get_grade(marks_dict[name]))

print("\n--- High Performers ---")

for name in marks_dict:
```

```
Marks Dict: {'Rahul': 85, 'Sneha': 92, 'Arjun': 78, 'Priya': 88, 'Karan': 70}
Attendance Dict: {'Rahul': 88.0, 'Sneha': 96.0, 'Arjun': 80.0, 'Priya': 92.0, 'Karan': 72.0}

--- Pass/Fail ---
Rahul Pass
Sneha Pass
Arjun Pass
Priya Pass
Karan Pass
```

```python

    print("\n--- Grades ---")

    for name in marks_dict:
        print(name, "→", get_grade(marks_dict[name]))

    print("\n--- High Performers ---")

    for name in marks_dict:
        if marks_dict[name] > 80 and attendance_percent[name] > 85:
            print(name)


    # ----------------------------------
    # PART 7 - COMBINED ANALYSIS
    # ----------------------------------

    print("\n--- Combined Data ---")
```

```
--- Grades ---
Rahul → B
Sneha → A
Arjun → B
Priya → B
Karan → C

--- High Performers ---
Rahul
```

```python
# PART 7 - COMBINED ANALYSIS
# ----------------------------------

print("\n--- Combined Data ---")

final_data = {}

for s in marks_data:
    name = s["name"]

    final_data[name] = {
        "marks": s["marks"],
        "attendance": attendance_percent[name],
        "course": s["course"],
        "grade": get_grade(s["marks"])
    }

for name, data in final_data.items():
```

```
--- Combined Data ---
Rahul {'marks': 85, 'attendance': 88.0, 'course': 'Python', 'grade': 'B'}
Sneha {'marks': 92, 'attendance': 96.0, 'course': 'Data Engineering', 'grade': 'A'}
Arjun {'marks': 78, 'attendance': 80.0, 'course': 'Python', 'grade': 'B'}
Priya {'marks': 88, 'attendance': 92.0, 'course': 'AI', 'grade': 'B'}
Karan {'marks': 70, 'attendance': 72.0, 'course': 'Python', 'grade': 'C'}

--- Final Result ---
Topper: Sneha
```

```python


# ----------------------------------
# PART 8 - FILE OUTPUT
# ----------------------------------

with open("report.txt", "w", encoding="utf-8") as f:
    f.write("Student Report\n")
    for name, data in final_data.items():
        f.write(f"{name} - Marks: {data['marks']} - Attendance: {round(data['attendance'],1)}% - Grade: {data['grade']}\n")


with open("eligible_students.txt", "w") as f:
    for name in eligible:
        f.write(name + "\n")


# ----------------------------------
```

```
Karan {'marks': 70, 'attendance': 72.0, 'course': 'Python', 'grade': 'C'}

--- Final Result ---
Topper: Sneha
Best Attendance: Sneha
Average Marks: 82.6
Eligible Students: Rahul, Sneha, Arjun, Priya
Students Needing Improvement: Karan

Process finished with exit code 0
```

```python
# ----------------------------------
# FINAL OUTPUT
# ----------------------------------

print("\n--- Final Result ---")

print("Topper:", topper["name"])
print("Best Attendance:", best_attendance)
print("Average Marks:", round(avg_marks, 1))
print("Eligible Students:", ", ".join(eligible))
print("Students Needing Improvement:", ", ".join(improvement))
```

```
--- Final Result ---
Topper: Sneha
Best Attendance: Sneha
Average Marks: 82.6
Eligible Students: Rahul, Sneha, Arjun, Priya
Students Needing Improvement: Karan

Process finished with exit code 0
```